Generative AI (GenAI)

Generative Artificial Intelligence (Generative AI) is a branch of Artificial Intelligence that can create new content such as text, images, code, audio, and videos by learning patterns from existing data.

Examples: ChatGPT, Google Gemini, DALL·E, and GitHub Copilot.

How Generative AI Works

Input / Prompt

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Data Processing

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Trained AI Model

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Pattern Recognition

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Content Generation

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Final Output

1). **Input / Prompt:** The user provides instructions or a question.

2). **Data Processing:** The input is converted into a suitable format for the AI model.

3). **Trained AI Model:** The model uses patterns learned during training.

4). **Pattern Recognition:** The model identifies relevant patterns and relationships.

5). **Content Generation:** The model generates new content based on the input.

6). **Final Output:** The system produces text, images, code, audio, or video.

Types of Generative AI

A. Text Generation

Generates human-like text based on user instructions.

Example: ChatGPT generates an answer to a question or creates study notes.

B. Image Generation

Creates new images from text descriptions or other inputs.

Example: DALL·E generates an image from the prompt "A robot studying in a library."

C. Code Generation

Generates computer programs and code based on instructions.

Example: GitHub Copilot suggests Python code for a program.

D. Audio Generation

Creates speech, music, and other sounds.

Example: AI generates spoken audio from written text.

E. Video Generation

Creates video clips from text prompts, images, or other inputs.

Example: AI generates a short video of a person walking through a forest.

Technologies Used in Generative AI

1). **Generative Adversarial Networks (GANs):** Use two neural networks, a generator and a discriminator, to generate realistic data.

2). **Variational Autoencoders (VAEs):** Learn data patterns and generate new samples from a learned latent representation.

3). **Transformers:** Use attention mechanisms to learn relationships between data elements and generate text, code, and other content.

4). **Diffusion Models:** Generate images and other data by gradually removing noise from a noisy input.

Applications of Generative AI

1). **Education:** Generates notes, quizzes, summaries, and explanations.

2). **Healthcare:** Helps generate synthetic medical data and supports medical research.

3). **Software Development:** Generates code and assists in debugging.

4). **Content Creation:** Creates articles, images, videos, and advertisements.

5). **Entertainment:** Generates music, stories, and visual effects.

6). **Customer Support:** Generates responses through AI chatbots.

Advantages of Generative AI

1). Saves time by automating content creation.

2). Generates new content quickly.

3). Improves productivity in education and software development.

4). Supports creativity and idea generation.

5). Can provide personalized learning materials.

Limitations of Generative AI

1). May generate incorrect or misleading information.

2). Can produce biased content due to biases in training data.

3). May raise copyright and privacy concerns.

4). Requires computing resources for training and operation.

5). Generated content may require human verification.

Generative AI is an important branch of Artificial Intelligence that creates new content by learning patterns from existing data. It is widely used for text, image, code, audio, and video generation in education, healthcare, software development, and entertainment.

Large Language Models (LLMs)

A Large Language Model (LLM) is a type of Artificial Intelligence model trained on a large amount of text data to understand and generate human language.

LLMs can answer questions, summarize text, translate languages, write content, and generate computer code. **Examples:** ChatGPT, Google Gemini, and Claude.

How LLMs Work

LLMs learn patterns, relationships, and context from text data. They use these learned patterns to predict and generate the next token (a word or part of a word).

Example:

Input: “Artificial Intelligence is a…”

Possible prediction: “technology”

The model generates text one token at a time until it completes the response.

How LLMs Are Trained

Large Text Dataset

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Data Preprocessing

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Tokenization

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Model Initialization

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Model Training

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Error Calculation

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Weight Adjustment

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Trained LLM

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Text Generation

Steps of Training

1). **Data Collection:** A large amount of text is collected from sources such as books, articles, websites, and code repositories.
2). **Data Preprocessing:** The collected data is cleaned to remove unwanted content, duplicates, and formatting problems.
3). **Tokenization:** The text is divided into smaller units called tokens. These tokens are converted into numerical representations that the model can process.
4). **Model Initialization:** The model is initialized with parameters called weights, which are adjusted during training.
5). **Model Training:** The model learns patterns in the text by predicting missing or upcoming tokens. Many LLMs use the Transformer architecture.

6). **Error Calculation:** The predicted tokens are compared with the expected tokens, and a loss function measures the prediction error.
7). **Weight Adjustment:** An optimization algorithm, commonly using gradient descent and backpropagation, adjusts the model's weights to reduce prediction error.
8). **Fine-Tuning:** The pretrained model may be further trained on specific datasets or human feedback to improve its ability to follow instructions and produce useful responses.
9). **Text Generation:** After training, the model receives a prompt and generates a response based on patterns learned during training.

Transformer Architecture

The Transformer is a neural network architecture widely used to build modern LLMs.

Its important components include:

1). **Token Embeddings:** Convert tokens into numerical vectors.

2). **Positional Information:** Helps the model understand the order of tokens.

3). **Attention Mechanism:** Helps the model identify relationships between tokens and focus on relevant context.

4). **Neural Netwaork Layers:** Process information and learn complex language patterns.

5). **Output Layer:** Produces scores used to predict the next token.

Applications of LLMs

1). **Chatbots:** Answer user questions and maintain conversations.

2). **Education:** Explain concepts and generate study notes.

3). **Text Summarization:** Convert long documents into shorter summaries.

4). **Machine Translation:** Translate text between languages.

5). **Code Generation:** Generate and explain computer programs.

6). **Content Creation:** Write articles, emails, and reports.

7). **Question Answering:** Provide responses to natural-language questions.

Advantages of LLMs

1). Understand and generate human-like text.

2). Perform multiple language-related tasks.

3). Process and summarize large amounts of text.

4). Support learning, research, and software development.

5). Generate responses quickly.

Limitations of LLMs

1). May generate incorrect or misleading information.

2). Can produce biased responses due to training data.

3). Require significant computing resources for training.

4). May not always understand context correctly.

5). Can generate confident answers without reliable factual support.

Large Language Models are AI models trained on large amounts of text to learn language patterns and generate responses. They commonly use Transformer architecture and are trained through token prediction, error calculation, and weight adjustment. LLMs are widely used in chatbots, education, translation, summarization, and code generation.

Popular Large Language Models (LLMs)

1. ChatGPT (OpenAI)

ChatGPT is an AI chatbot developed by OpenAI. It uses Large Language Models (LLMs) to understand user instructions and generate human-like responses in natural language.

ChatGPT can perform tasks such as answering questions, writing content, explaining concepts, generating computer programs, and analysing information.

Working of ChatGPT:

1). **User Input:** The user enters a question or instruction called a prompt.

2). **Tokenization:** The input text is divided into smaller units called tokens.

3). **Model Processing:** The model processes the tokens and uses learned language patterns and context to understand the request.

4). **Response Generation:** The model predicts and generates the response one token at a time.

5). **Output:** The generated response is displayed to the user.

Applications:

- **Education:** Explains difficult topics and prepares study notes.
- **Programming:** Generates, explains, and helps debug code.
- **Content Writing:** Creates articles, emails, reports, and summaries.
- **Question Answering:** Answers questions in natural language.
- **Problem-Solving:** Helps solve mathematical and logical problems.

Example: A student enters the prompt, "Explain Artificial Intelligence with examples." ChatGPT generates an explanation of AI with relevant examples.

2. Gemini (Google)

Gemini is a family of AI models developed by Google. It can understand and generate text and, depending on the model and available features, process other types of information such as images, audio, video, and documents.

Gemini supports multimodal AI, which means that an AI system can work with more than one type of input or output.

Working of Gemini:

1). **Input:** The user provides text, an image, a document, or another supported type of input.

2). **Input Processing:** The system converts the input into a suitable representation for the model.

3). **Context Analysis:** The model analyses the input and identifies relevant patterns and relationships.

4). **Model Processing:** The model uses its learned knowledge to understand the request and prepare a response.

5). **Output Generation:** The system generates a response in a suitable format, such as text or other supported content.

Applications:

- **Research:** Helps find and organize information.
- **Image Analysis:** Explains the contents of supported images.
- **Document Analysis:** Summarizes and analyses documents.
- **Education:** Explains concepts and helps solve questions.
- **Content Generation:** Creates text and other supported content.
- **Productivity:** Helps with planning, writing, and organizing tasks.

Example: A student uploads an image of a mathematical problem and asks Gemini to solve it. The system analyses the image and generates a step-by-step solution.

3. Claude (Anthropic)

Claude is a family of AI models developed by Anthropic. It is designed for tasks involving language understanding, reasoning, writing, coding, and document analysis.

Claude can process user instructions, identify relationships between ideas, and generate responses based on the provided context.

Working of Claude:

1). **User Input:** The user provides a question, instruction, code, or document.

2). **Tokenization:** The input is converted into tokens that the model can process.

3). **Context Processing:** The model analyses the input and considers relationships between different parts of the text.

4). **Reasoning and Response Preparation:** The model processes the request and develops a suitable response.

5). **Output Generation:** The model generates text or code based on the input and learned patterns.

Applications:

- **Document Analysis:** Summarizes long documents and extracts important information.
- **Programming:** Writes, explains, and helps debug computer code.
- **Content Writing:** Prepares reports, articles, and professional emails.
- **Learning:** Explains complex subjects in simpler language.
- **Data Interpretation:** Helps interpret information provided in text or supported files.

Example: A programmer provides a Python program containing an error and asks Claude to identify the problem. Claude analyses the code and suggests a correction with an explanation.

4. Llama (Meta)

Llama is a family of Large Language Models developed by Meta. These models can perform language-related tasks such as text generation, question answering, summarization, and code generation.

Many Llama models have publicly available weights under specific licence terms. Developers can use eligible models to build and customize AI applications according to the applicable licence.

Working of Llama:

1). **Input:** The user provides a prompt, question, or text.

2). **Tokenization:** The input is divided into tokens.

3). **Model Processing:** The model processes the tokens using its learned parameters and language patterns.

4). **Token Prediction:** The model predicts the next token based on the preceding context.

5). **Response Generation:** The process repeats until the model produces a complete response.

Applications:

- **AI Chatbots:** Developers can build conversational AI systems.
- **Text Generation:** Generates descriptions, articles, and summaries.
- **Code Generation:** Helps create and explain computer programs.
- **Custom AI Applications:** Supports the development of specialized language-based tools.
- **Research:** Enables experimentation with language models and AI applications.

Example: A developer uses an eligible Llama model to create a chatbot for a college website that answers questions about courses, admissions, and college facilities.

Comparison of Popular LLMs

Feature	**ChatGPT**	**Gemini**	**Claude**	**Llama**
Developed by	OpenAI	Google	Anthropic	Meta
Type	AI assistant and chatbot	Multimodal AI model family	AI assistant and model family	LLM model family

Main uses	Writing, coding, learning, and Q&A	Research, multimodal tasks, and productivity	Writing, coding, reasoning, and document analysis	Language tasks and custom AI development
Input types	Text and other supported media	Text and other supported media	Text and other supported media	Depends on the model and implementation
Customization	Depends on available tools and plans	Depends on available tools and plans	Depends on available tools and plans	Eligible models can be customized under their licences
Access	Web, apps, and APIs	Web, apps, and APIs	Web, apps, and APIs	Model downloads and supported platforms

Example of the Same Task Using Different LLMs

Task: Generate a Python program to calculate the sum of two numbers.

- **ChatGPT:** Can generate the Python program and explain each line.
- **Gemini:** Can generate the program and help explain or analyse related information.
- **Claude:** Can generate the program and explain its logic or suggest improvements.
- **Llama:** Can generate the program when used through a suitable model or application.

All four can perform similar tasks, but their responses, capabilities, accuracy, and performance may differ depending on the model version and configuration.

Tokens, Context Window, and Temperature

1. Tokens

Tokens are small units of text that an AI model processes. A token can be a word, part of a word, or a punctuation mark.

Example:

Input: Artificial Intelligence is useful.

The text is divided into smaller units called tokens. The exact token division depends on the model's tokenizer.

Input Text

↓

"Artificial Intelligence is useful."

↓

Tokenization

↓

[Token 1] [Token 2] [Token 3] ...

↓

AI Model Processing

How Tokens Work

1). **Text Input:** The user enters a sentence or question.

2). **Tokenization:** The text is divided into tokens.

3). **Numerical Representation:** The tokens are mapped to numerical IDs that the model can process.

4). **Model Processing:** The AI model analyses relationships between the tokens.

5). **Output Generation:** The model predicts and generates new tokens to form a response.

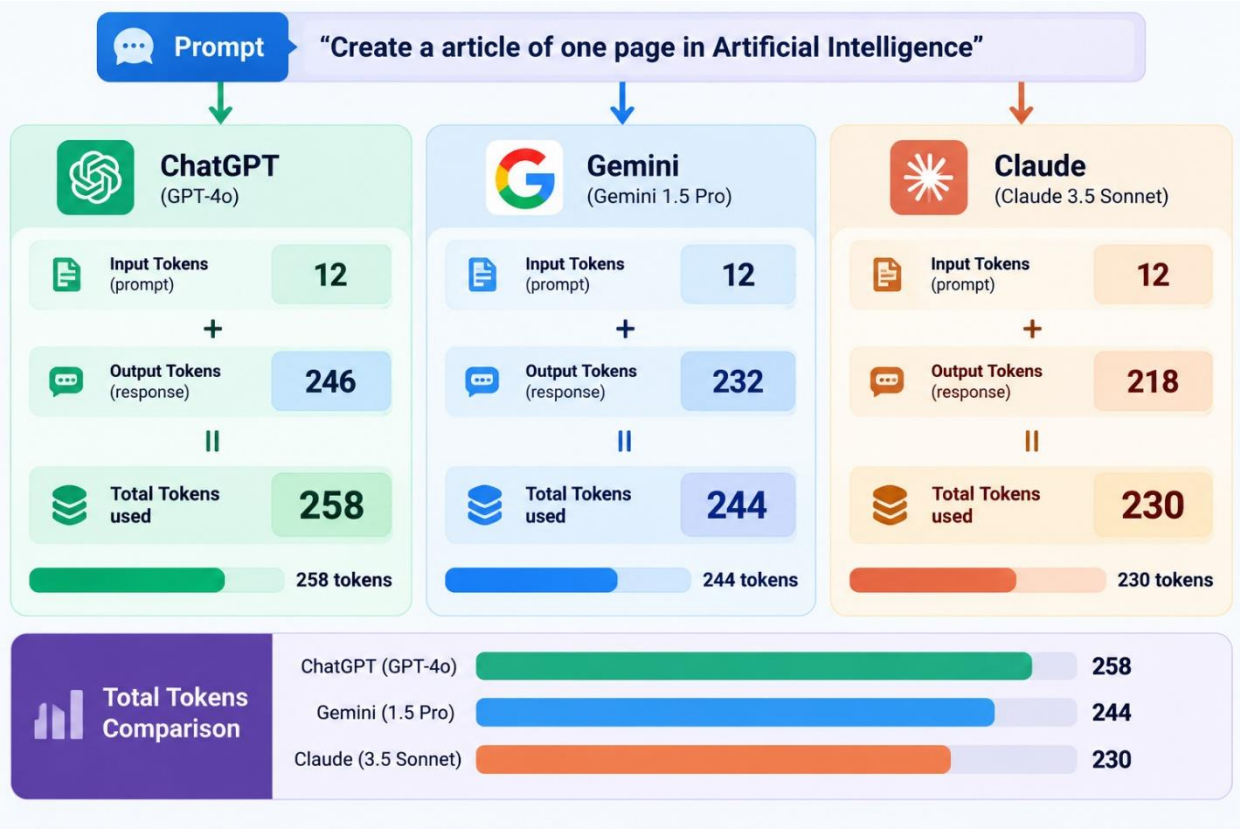
6). **Text Conversion:** The generated tokens are converted back into readable text.

Uses of Tokens:

1). Help AI models process text.

2). Measure the amount of text given to or generated by a model.

3). Help determine the processing cost of many AI services.



2. Context Window

The context window is the maximum amount of information, measured in tokens, that an AI model can consider during a single interaction.

It may include the current prompt, previous conversation messages, and the response being generated, depending on the model.

Example:

Suppose an AI model has a context window of 10,000 tokens. The prompt, relevant conversation history, and generated response must fit within the model's applicable token limit.

How the Context Window Works

1). **Input Reception:** The model receives the user's question.

2). **Context Collection:** Relevant previous messages and instructions may be included.

3). **Token Counting:** The input is represented and measured in tokens.

4). **Limit Checking:** The system ensures that the input and requested output fit within the applicable limits.

5). **Response Generation:** The model generates a response using the available context.

Why Is the Context Window Important?

1). **Long Conversations:** A larger context window can accommodate more conversation history.

2). **Document Analysis:** It allows the model to process longer documents in a single interaction.

3). **Context Understanding:** It can help the model consider more relevant information when generating a response.

4). **Input and Output Limits:** It determines how much information can fit into a model interaction.

Example

A student provides a 100-page document and asks the AI to summarize it.

- If the document fits within the available context capacity, the model may process it in one interaction.

- If it exceeds the available capacity, the document may need to be divided into smaller sections.

A larger context window does not guarantee perfect understanding or recall of every detail.

Uses of Context Window:

1). Helps the model process long conversations.

2). Allows the model to analyse lengthy documents.

3). Determines how much information can be considered at one time.

3. Temperature

Temperature is a parameter that controls the randomness of an AI model's output when selecting the next token.

- **Low temperature:** Produces more predictable and consistent responses.
- **High temperature:** Produces more varied and potentially creative responses.

Example:

Prompt: "Write a sentence about a beautiful garden."

- **Low temperature:** "The garden is beautiful and full of flowers."
- **High temperature:** "A magical garden blooms with colourful flowers beneath the golden sunlight."

These are illustrative examples; actual responses depend on the model and settings.

No, temperature in Generative AI is not the actual environmental temperature. It has nothing to do with hot or cold weather.

It is a mathematical setting that controls how predictable or varied an AI model's responses are.

Why is it called Temperature?

The term comes from statistical physics. In AI, temperature is used in a mathematical formula to control the probability of selecting different words or tokens.

How Does It Work?

Suppose you ask an AI:

“The sun rises in the ____.”

The AI model calculates probabilities for possible next words.

Possible word	Original probability
East	90%
West	5%
Sky	5%

Low temperature (e.g., 0.2):

- The model strongly favours the most probable word.
- It is more likely to choose “East.”

High temperature (e.g., 1.5):

- The model distributes probability more evenly among possible words.
- Less probable words have a greater chance of being selected.

These percentages are illustrative; actual probabilities depend on the model and temperature setting.

Real-Life Analogy

Imagine choosing an answer in an exam:

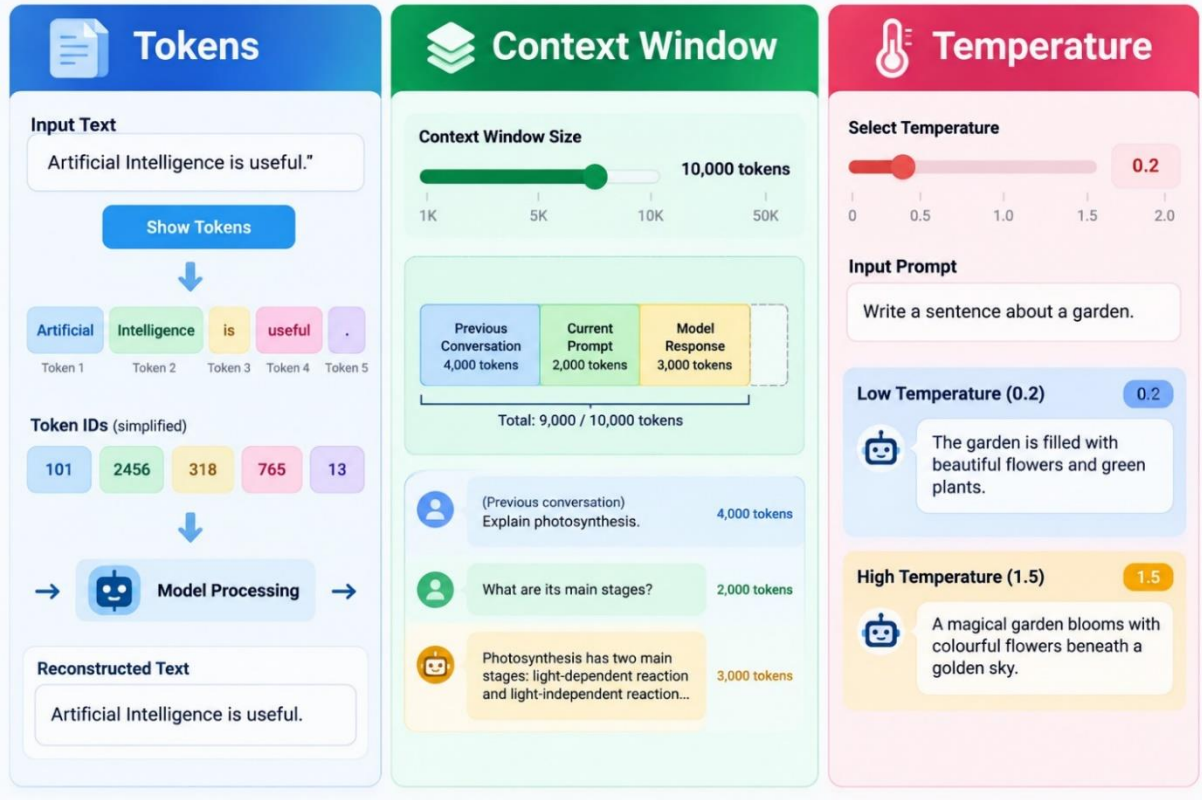
- **Low temperature:** You choose the answer you believe is most likely to be correct.
- **High temperature:** You are more willing to consider less likely alternatives.

Uses of Temperature:

1). Controls the variability of generated text.

2). Low temperature is useful for predictable responses.

3). Higher temperature can be useful for creative writing and brainstorming.



Difference Between Tokens, Context Window, and Temperature

Feature	**Tokens**	**Context Window**	**Temperature**
Meaning	Units of text processed by AI	Maximum token capacity for an interaction	Controls randomness in token selection

Main purpose	Represents text in a model-readable form	Determines how much information fits	Influences response variability
Example	Words or parts of words	A limit of 10,000 tokens	Low or high temperature
Effect	Affects text measurement and processing	Affects how much context can be considered	Affects predictability and creativity

Multimodal AI

Multimodal AI is a type of Artificial Intelligence that can process and understand multiple types of data, such as text, images, audio, and video, within a single AI system.

Examples: GPT-4o by OpenAI and Gemini by Google.

Types of Data Used in Multimodal AI

1). **Text:** Written words, sentences, and questions.

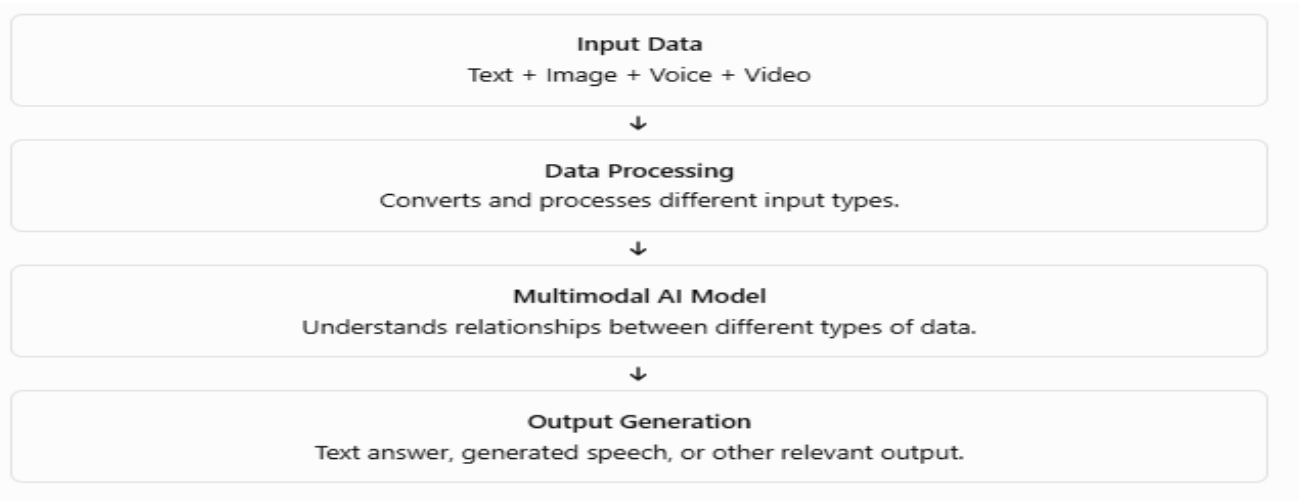
2). **Images:** Photographs, diagrams, and scanned documents.

3). **Audio:** Human speech, music, and other sounds.

4). **Video:** A sequence of images along with audio.

5). **Combined Data:** Two or more types of data processed together.

Working of Multimodal AI



Examples

- **GPT-4o:** Can understand text, analyze images, and interact through voice.
- **Gemini:** Can process and reason across different types of information, including text, images, audio, and video, depending on the model and application.

GPT-4o

GPT-4o is a multimodal AI model developed by OpenAI. The letter “o” stands for omni, meaning all or everything.

Its key concept is to work with multiple modalities, including text, images, and audio, within one model.

Working of GPT-4o:

1). Accepts text, images, or audio as input.

2). Processes the information and identifies relationships between the input types.

3). Understands the user's request using the combined information.

4). Generates an appropriate response, such as text or speech.

Example: A student shows GPT-4o a photograph of a mathematics question and asks a question using voice. The model can analyze the image, understand the spoken question, and explain the solution.

Key feature: GPT-4o supports natural, real-time voice interaction and multimodal understanding.

Gemini

Gemini is a family of multimodal AI models developed by Google. It is designed to understand and work with different types of information, including text, images, audio, and video, depending on the model and application.

Working of Gemini:

1). Receives information such as text, images, audio, or video.

2). Processes the available input modalities.

3). Identifies patterns and relationships among the information.

4). Generates a relevant answer or other supported output.

Example: A student uploads a science experiment video to Gemini and asks for an explanation. Gemini can use the visual information and available audio to help explain what is happening in the experiment.

Key feature: Gemini supports multimodal reasoning and can work with long or varied inputs, depending on the model's capabilities.

Difference Between GPT-4o and Gemini

Basis	**GPT-4o**	**Gemini**
Developer	OpenAI	Google
Type	Multimodal AI model	Family of multimodal AI models
Input	Text, images, audio, and supported modalities	Text, images, audio, video, and supported modalities
Main concept	Unified multimodal interaction, including real-time voice	Multimodal understanding and reasoning across different inputs
Example use	Discussing a diagram through voice interaction	Analyzing a video and answering questions about it

Applications of Multimodal AI

1). **Education:** Explaining diagrams and answering questions using text and images.

2). **Healthcare:** Analyzing medical images along with clinical information.

3). **Virtual Assistants:** Understanding spoken questions and responding to users.

4). **Autonomous Vehicles:** Processing camera images and other sensor data.

5). **Security:** Analyzing video footage and identifying objects.

6). **Content Creation:** Generating text, images, and audio.

Advantages

1). Processes multiple types of data in one system.

2). Provides richer and more context-aware responses.

3). Supports natural voice-based interaction.

4). Helps solve complex problems using information from different sources.

Limitations

1). Requires high computational power and memory.

2). Training and development can be expensive.

3). May produce incorrect interpretations or outputs.

4). Can raise privacy and security concerns when processing sensitive data.

Multimodal AI combines different types of data, such as text, images, audio, and video, to understand information more effectively and generate useful responses. It is an important technology used in modern AI assistants, healthcare, education, and automation.

GPT-4o – Demonstration

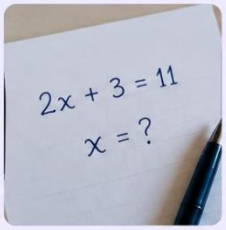
Multimodal: Text + Image + Voice + Video

1. Text Input → Image Response

Generate an image of a peaceful beach at sunset with palm trees.



2. Image Input → Text Response



To solve the equation:

2x + 3 = 11
2x = 11 – 3
2x = 8
x = 8 ÷ 2

x = 4

Text Input

Image Input

GPT-4o

Voice Input

Video Input

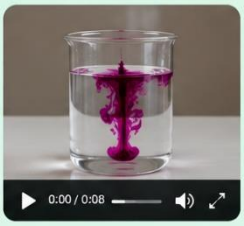
3. Voice Input → Voice Response

Explain photosynthesis in simple words.

GPT-4o

Photosynthesis is the process by which green plants make their own food using sunlight, water and carbon dioxide.

4. Video Input → Text Response



The video shows a drop of food colour added to water. The colour spreads through the water due to diffusion, where particles move from a region of higher concentration to a region of lower concentration until it mixes evenly.

Google Gemini – Demonstration

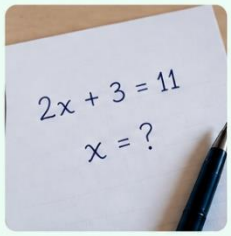
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Text Input

Image Input

Gemini

Voice Input

Video Input

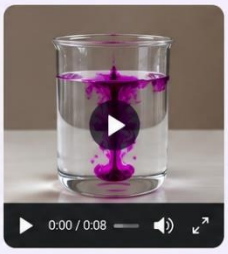
3. Voice Input → Voice Response

Explain photosynthesis in simple words.

Photosynthesis is the process by which green plants make their own food using sunlight, water and carbon dioxide.

0:00 / 0:10

4. Video Input → Text Response



The video shows a drop of food colour added to water. The colour spreads through the water due to diffusion, where particles move from a region of higher concentration to a region of lower concentration until it mixes evenly.

AI Image Generation — Stable Diffusion, Midjourney and DALL·E

AI Image Generation is a technique in Artificial Intelligence that creates new images from text descriptions (prompts), reference images, or other inputs using trained AI models.

For example, entering the prompt “A futuristic city with flying cars at sunset” can generate an image matching the description.

Working of AI Image Generation

Text Prompt

“A beautiful house near the mountains”

AI Model

Understands the prompt and generates an image.

Generated Image

A house surrounded by mountains.

Popular AI Image Generation Tools

Stable Diffusion

- An AI image-generation model developed by Stability AI and its ecosystem.
- Uses diffusion techniques to generate images from text prompts.
- Offers customization and can be run locally with suitable hardware.

Midjourney

- An AI image-generation service known for artistic, detailed, and visually appealing images.
- Generates images from text prompts.
- Commonly used for digital art, concept art, and creative designs.

DALL·E

- An AI image-generation model developed by OpenAI.
- Generates images from natural-language descriptions.
- Useful for illustrations, creative concepts, and visual content.

Demo

Prompt: “A futuristic city with flying cars at night.”

Expected process:

1). Enter the prompt into an image-generation tool.

2). The AI interprets the description.

3). The model generates an image matching the prompt.

4). Review the result and modify the prompt if necessary.

Applications

1). Digital art and graphic design.

2). Advertising and marketing.

3). Education and visual learning.

4). Game and movie concept design.

5). Product visualization.

6). Social media content creation.

Advantages

- Generates images quickly from text descriptions.
- Helps users create visuals without advanced drawing skills.

- Supports creative experimentation.
- Reduces the time needed to produce initial design concepts.

Limitations

- May generate inaccurate details or distorted objects.
- Results depend on the prompt and model.
- Requires computational resources or access to an online service.
- May raise copyright, privacy, and ethical concerns.

AI image generation uses trained AI models to create images from text prompts and other supported inputs. Stable Diffusion, Midjourney, and DALL·E are popular examples that support creativity and visual content creation across many fields.

